

INTERNAL · CONFIDENTIAL

Santander Business Banking

Concept Prototype — Technical Report

A front-end exploration of paperless workflows for business banking

Field	Value
Author	Alan Davidson
Prepared	July 2026
Version	1.0
Status	Prototype — internal discussion

Section 1

1 Executive summary

This report describes a self-initiated concept prototype for Santander Business Banking. The prototype is a front-end only application that demonstrates how paper-based business-banking processes — mandate changes, bulk payments, account closures, disputes, cross-border onboarding and more — can be delivered as guided, self-serve, digitally-signed workflows inside the banking app.

It is a demonstration piece for internal discussion and customer research. It has no backend, uses no real customer data, and makes no live system connections. Its value is as a tangible, clickable articulation of a paperless service model and the regulatory controls that model must respect.

Dimension	Summary
Workflows	17 end-to-end paperless journeys
Entity types	7 business types (sole trader to society)
Platform	React single-file front end, one self-contained HTML bundle
Scope	Front-end concept only — no backend, no real data

Section 2

2 Background & problem statement

Many business-banking service events still depend on paper: posted mandate-variation forms, wet-ink signatures, branch visits to reset access, faxed international payment details, and written requests for balance certificates that take days to arrive. Each step adds delay, cost and operational risk.

The prototype asks: what would these processes look like re-imagined as in-app, self-serve, compliant-by-design workflows — with the regulatory checks built into the flow rather than bolted on with paper?

Section 3

3 Objectives & scope

Objectives

- Demonstrate paperless equivalents for the most paper-heavy business-banking service events.
- Show the regulatory controls each workflow must satisfy, expressed inside the user experience.
- Provide a clickable artefact for internal review and customer research.
- Explore a consistent design language and accessibility standard.

Non-goals (out of scope)

- No backend, database or integration with live banking systems.
- No real customer, account or payment data — all figures are illustrative.
- No production security hardening.
- No processing of real payments, identity documents or regulatory submissions.

Section 4

4 System architecture

The prototype is built as a single React component in one file, compiled by Vite into a single self-contained HTML bundle. This keeps the whole product readable top-to-bottom and trivial to deploy or share.

- Single-file React front end; all state held at the top of one component.
- Compiled to one self-contained HTML artefact.
- Two shells — mobile and desktop — sharing all state.
- A workflow-overlay model: each journey is a full-screen, multi-step wizard.
- An entity system that reconfigures accounts, mandate rules and copy across 7 business types.

Mandate & authorisation model

Each account carries a signing rule (any-one, any-two, or all). Workflows resolve the strictest applicable rule across the accounts involved, determining whether co-signing or a cooling-off period is required.

Section 5

5 Workflow catalogue

The prototype implements 17 end-to-end workflows, each replacing a specific paper or branch-based process.

Workflow	What it does	Basis
Account closure	In-credit & cooling-off checks	BCOBS
Business details	Name/address/contact + proof	KYB
Mandate changes	Add/remove signatories, signing rule	SMR
Bulk payments	Payee book, CoP, scheduled runs	CoP
Pre-approved lending	Offer, calculator, draw-down	CCA 1974
International FX	Rate & fee disclosure	Cross-border
Dormant reactivation	Reactivate or close	Dormancy
Personal/business unlink	Separate personal data	GDPR
Credit ring-fence	Exclude personal data from credit	GDPR 5(1)(c)
Signatory ID register	KYC/KYB evidence lists	MLR 2017
MTD VAT submission	Categorise & submit VAT	HMRC MTD
Complaint handling	Intake to FOS escalation	FCA DISP
Standing orders & DDs	Set up SO; cancel DD	DD Guarantee
Transaction dispute	Chargeback/fraud/DD routes	PSR 2017
Intl. beneficiary	Screen & verify a payee	MLR/sanctions
Balance certificate	Sealed certificate/reference	Evidence
Trusted devices	Devices, activity, sign-out	PSD2 SCA

Section 6

6 Regulatory & compliance mapping

The workflows are designed around the controls a real implementation would need, surfaced in the experience so reviewers can see where each obligation is met.

Framework	How the prototype reflects it
SCA / PSD2 RTS	Strong authentication and step-up before sensitive actions
Confirmation of Payee	Account-name verification before first payment

Direct Debit Guarantee	Immediate refund route for DD disputes
PSR 2017	Unauthorised-transaction refund and chargeback handling
FCA BCOBS / Consumer Duty	Cooling-off and fair treatment on changes
FCA DISP	Complaint intake, decision and Ombudsman escalation
FCA SYSC 9	Actions recorded to an audit trail
MLR 2017	Sanctions, PEP and adverse-media screening
HMRC MTD	Digital VAT return preparation and submission
GDPR	Data separation, ring-fencing, purpose limitation

Section 7

7 Engineering & quality approach

The prototype is developed through a repeatable pipeline that separates generation, review, gating and runtime verification:

- Scaffold — new workflows generated from a template that prevents recurring defects.
- Review — parallel multi-lens review of architecture, security, engineering, design and state.
- Gate — a pre-commit check blocks accessibility, colour-token, markup and build regressions.
- Runtime verification — the built app is driven headlessly through real journeys with an automated WCAG audit.

Design system

A consistent visual language governs spacing, colour, typography and interaction: a defined spacing scale, the Santander brand red for primary actions, a warm neutral palette, tabular figures for monetary values, and one primary call-to-action per screen.

Section 8

8 Accessibility

Accessibility is treated as a build gate. Interactive controls carry visible focus states and accessible names; status is never conveyed by colour alone; and the interface is audited against WCAG 2.1 AA using an automated engine as part of the verification run. The core screens pass the automated colour-contrast and name/role checks.

Section 9

9 Limitations

- Front-end only: no data is stored, transmitted or processed.
- All names, figures and references are illustrative and fictional.
- Regulatory behaviours are represented for demonstration, not certified for production.
- Not security-tested to production standards.

Section 10

10 Indicative next steps

- Prioritise the highest-value journeys for a backend-integrated pilot.
- Validate the flows with business customers through structured research.

- Complete a full accessibility conformance review with assistive-technology testing.
- Define integration, security and data-protection requirements for a production build.

Prototype only — no backend, no real data, no live system connections. Prepared for internal discussion.